Module 4: Complete Analysis Report.

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CS4412-DataMining.

INTRODUCTION

- *Overview.*

This report presents the complete data analysis performed on the UCI Online Retail dataset on the UCI dataset repository. The dataset captures transactional data from a UK-based online retailer spanning the period of December 2010 to December 2011. The data contained 541,909 raw transactions across eight attributes. The transactions were then limited to the United Kingdom which then focused on 485,123 transactions from 3,920 unique United Kingdom customers over 18,019 invoices and 4,007 products.

- *Discovery Questions.*

Upon visualizing the data, three discovery questions were defined to guide the analytical decisions made throughout the analysis.

  - What products are frequently purchased together?
  - What natural customer segments exist within the data?
  - What distinct purchasing patterns emerge among the segments by season?

These questions are approached and answered throughout the data analysis process. Question 1 is answered through Association rule mining, Question 2 is answered through an RFM based customer segmentation analysis, and Question 3 answered through seasonal cross tabulation.

The overarching goal of this project was to gain actionable business data which could inform the business practices of the online retailer.

METHODOLOGY.

- *Data Cleaning and Preprocessing.*

The raw dataset contained some imperfections which could distort the analysis process. Thus, the following preprocessing steps were performed before running any algorithmic analysis on the data.

  - Cancellations were removed from the dataset.
  - The dataset was then restricted to the geographical region of the United Kingdom.
  - Rows with negative or zero quantities and unit prices were also dropped.
  - Rows with null or empty product descriptions were also excluded.
  - The invoice dates were parsed to datetime.
  - Month and total price columns were added to the dataset to help with subsequent analysis.

- Customers without valid customer IDs were removed from the dataset.

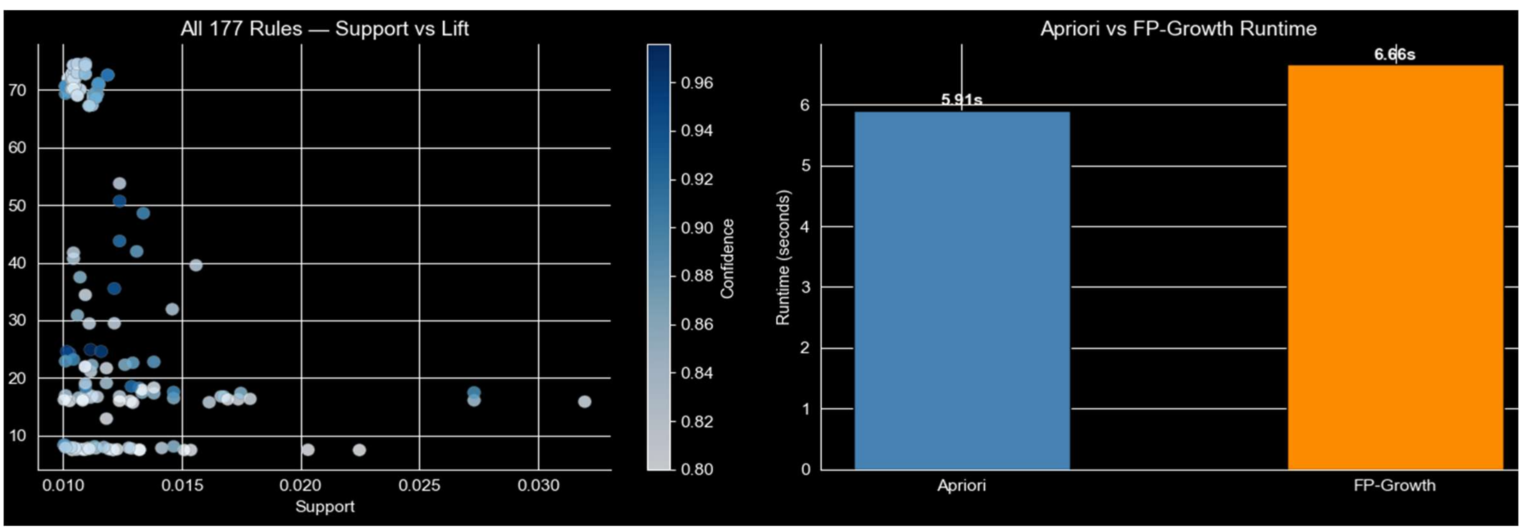
After the preprocessing the dataset was reduced to 485,123 valid transactions, spanned the period of December 01,2010 – December 09,2011, 3,920 unique customers with 18,019 invoices.

- *Association Rule Mining – Apriori & FP- Growth.*

The two algorithms were run at identical thresholds where the min_support = 0.01, and a min_confidence = 0.80. They were run on same basket matrix which contained 18,019 invoices and 4,007 unique products. The matrix was configured to be sparse to ensure that enough memory could be allocated during its construction. A denser matrix caused the program to terminate prematurely. The algorithms were also tested on a min_support = 0.02 where only five rules were generated

METRIC	APRIORI	FP-GROWTH.
FREQUENT ITEMSET	Same	Same
NUMBER OF RULES GENERATED	177	177
TIME TAKEN	~5.9s	~6.6s

Table 1 Showing the similarities and runtime comparison between Apriori and FP-Growth.



Note: Runtimes may be different due to device hardware configuration.

Figure 1 Showing item rules and algorithmic runtimes comparison.

- *Recency, Frequency, Monetary (RFM) Feature Engineering.*

Each customer with a valid CustomerID was reduced to three unique behavioral characteristics:

- **RECENCY**, which showed the number of days since last purchase using December 10,2011 as a date of reference.
- **FREQUENCY**, highlighting the number of invoices per customer.
- **MONETARY,** total amount spent in GBP.

- *Clustering – K-Means & Hierarchical Clustering.*

Both the K-Means and Hierarchical Clustering algorithms were run at a K=4 and on standardized RFM features. The best K value was 2 but the value K = 4 was selected to ensure that a reasonable number of actionable segments are present, the value was also selected due to its position at the elbow of the graph.

Three internal metrics were computed to assess the results: Silhouette (higher = better), Davies-Bouldin (lower = better), and Calinski-Harabasz (higher = better). Labels were then assigned to the groups that arose from the RFM centroid analysis.

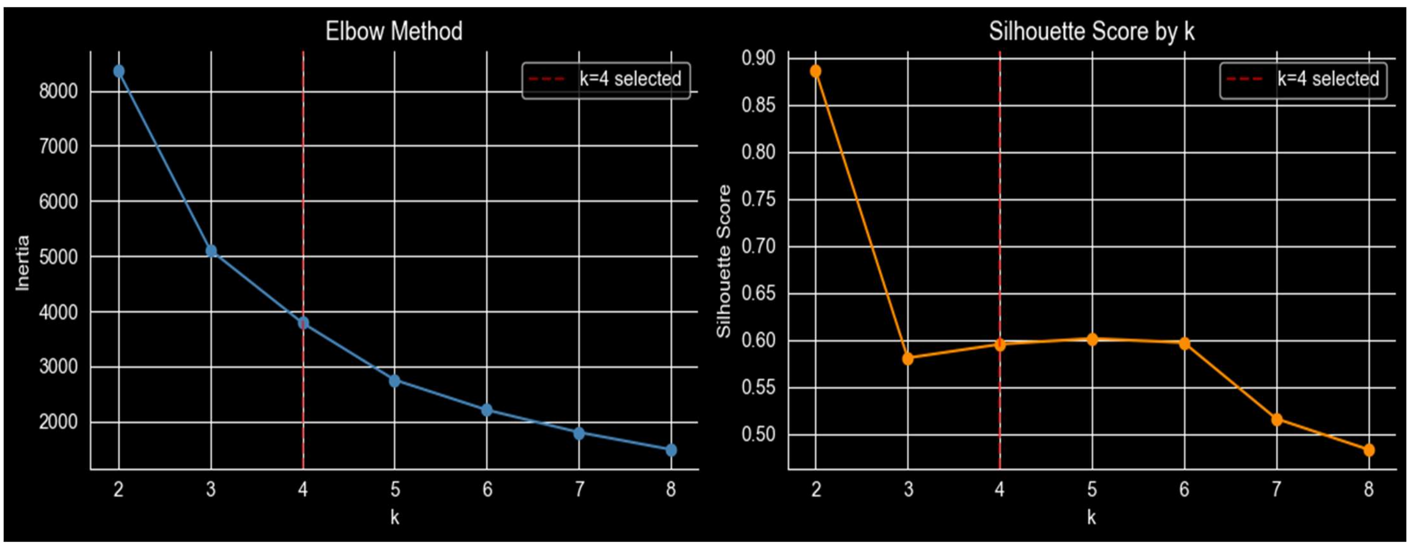


Figure 2 Showing K-Means selection by the elbow test.

- *Cluster Stability & K-means sensitivity test.*

To address any issues that may arise as per the validity of the Champions cluster, K-means was re-run with five random seeds: 0, 7, 13, 42, 99 to confirm that the cluster size is not seed dependent. A K value sensitivity test was conducted with K values of 3, 4, and 5 to assess whether an alternative value produces more balanced segmentation.

CENTROID	ASSIGNED LABEL	REASON.
0	At-Risk/ Lapsed	Highest average recency
1	Loyal regulars	Highest average frequency
2	New / Occasional	Unimpressionable behavior.
3	Champions	Highest average monetary spend

Table 2 Cluster labeling and Reasoning.

- *Classification – Decision Tree & Gaussian Naïve Bayes.*

The K-means cluster assignments were then used as class labels. A decision tree with max depth = 4 and random state=42 and Gaussian Naïve Bayes were implemented to highlight and identify which RFM feature/s distinguish each customer segment. The depth of the tree was limited to maximize interpretability of the results. They were then evaluated via 5-fold stratified cross-validation reporting macro-F1 scores.

- *Anomaly Detection – DBSCAN & LOF.*

DBSCAN with an epsilon = 0.5 and minimum samples = 5 and Local Outlier Factor with nearest neighbors = 20 and contamination = 0.05 were independently applied to the standardized RFM features. Any customer flagged by both methods simultaneously was identified and treated as high-confidence anomalies.

- *PCA Analysis.*

Principal components 1 and 2 were computed to assess the total variance captured by the RFM features.

- *Seasonal Analysis*

Calendar months were mapped to meteorological seasons: Winter: December – February; Spring: March – May; Summer: June – August; Autumn: September – November. Revenue and invoice counts were then cross tabulated against the four customer segments to analyze seasonal patterns which may exist amongst them.

RESULTS.

- *Three Co-Purchase Family Groups - Association Rule Mining Results.*

At min_support = 0.01 and confidence =0.08, both Apriori and FP-Growth produced exactly 177 rules. Three dominant product families emerged from the analysis:

PRODUCT FAMILY	RULES	AVERAGE CONFIDENCE	AVERAGE LIFT
HERB MARKERS	81	0.88	71.24
JUMBO BAG COLLECTION (RED RETROSPOT AND OTHERS)	36	0.84	9.5

REGENCY TEACUP SET(PINK/GREEN/ROSES)	20	0.87	26.8

Table 3 Showing the results of Association Mining Analysis.

- *Customer Segmentation Results – Clustering, Classification, RFM engineering results, and Anomaly Analysis.*

RFM engineered features were run through the K-Means algorithm at K = 4. This produced four customer segments with the following attributes:

SEGMENT	NUMBER(N)	% BASE	AVG RECENCY	AVG FREQUENCY	AVG MONETARY (£)
CHAMPIONS	3	0.1%	3.3	36.0	**207,560.2**
LOYAL REGULARS	52	1.3%	12.2	**43.8**	31172.2
AT-RISK/ LAPSED	989	25.2%	**244.6**	1.6	526.7
NEW / OCCASIONAL	2,876	**73.4%**	41.4	4.4	1579.9

Table 4 Customer Segmentation and Characteristic Attributes.

Note: Avg Recency = mean number of days since last purchase, Avg Frequency = mean number of invoices, and Avg Monetary = mean amount of spend.

With K-Means labels serving as a basis, classification of the customer base was evaluated on the three RFM features via a 5-fold stratified cross-validation:

CLASSIFIER	5-FOLD MACRO-F1	TRAINING ACCURACY	PRIMARY VALUE
DECISION TREE (DEPTH = 4)	0.842 +/- 0.057	0.974	Interpretable if-then rules; Gini importances.
GAUSSIAN NAÏVE BAYES	0.749 +/- 0.095	0.950	Probabilistic feature class associations.

Table 5 Showing the comparative results of classification.

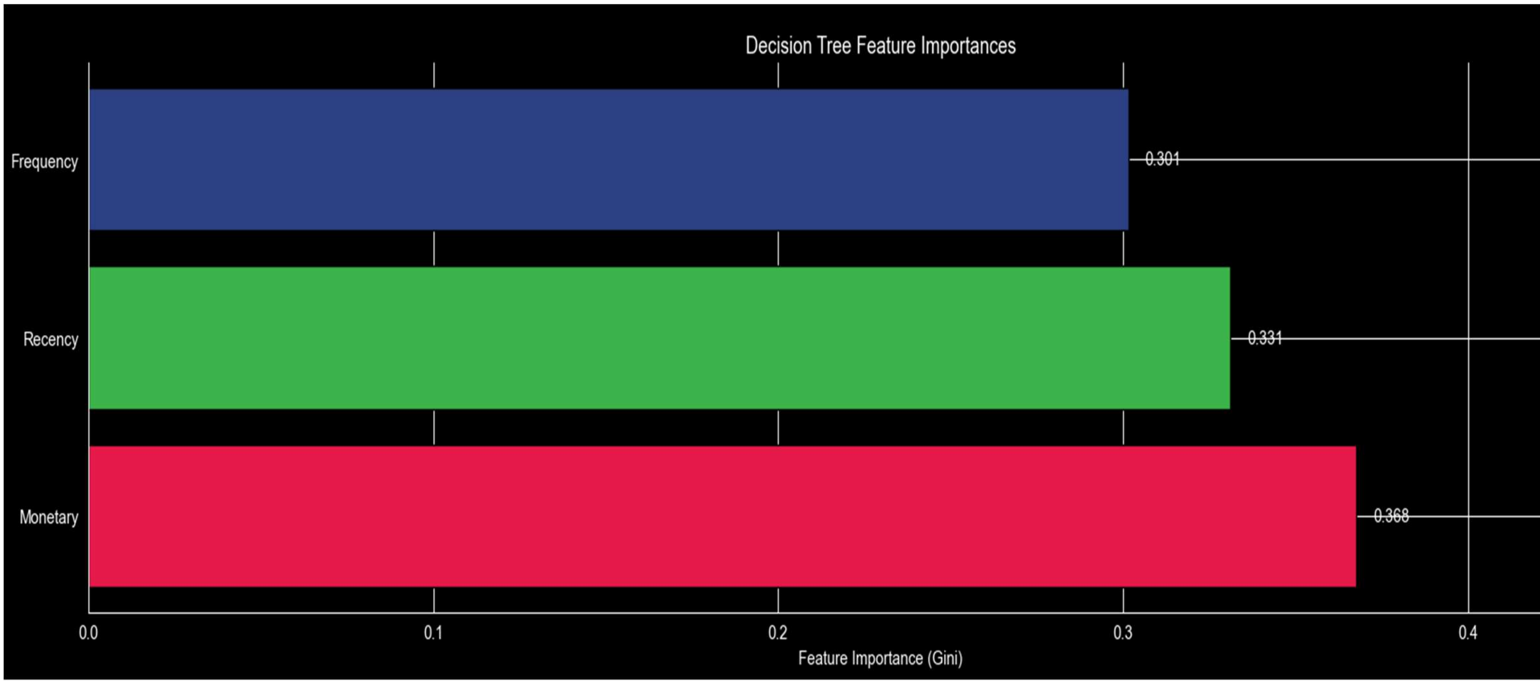


Figure 3 Showing feature importance comparison.

Per-Class Classification Reports

Decision Tree (depth = 4)

SEGMENT	PRECISION	RECALL	F1-SCORE	SUPPORT
AT-RISK / LAPSED	0.98	1.00	0.99	989
CHAMPIONS	1.00	1.00	1.00	3
LOYAL REGULARS	0.40	0.98	0.56	52
NEW / OCCASIONAL	1.00	0.97	0.98	2876

Table 6 Showing full decision tree classification report.

Gaussian Naïve Bayes

SEGMENT	PRECISION	RECALL	F1-SCORE	SUPPORT
AT-RISK / LAPSED	0.90	0.99	0.94	989
CHAMPIONS	1.00	1.00	1.00	3
LOYAL REGULARS	0.41	1.00	0.58	52
NEW / OCCASIONAL	1.00	0.93	0.97	2876

Table 7 Showing full Gaussian Naive Bayes classification report

To address any anomalies which may exist within the data, anomaly finding algorithms DBSCAN and LOF were used to process the data which resulted in the following:

DBSCAN noise points: 52 (1.3%), LOF anomalies:196 (5.0%), Flagged by both:38.

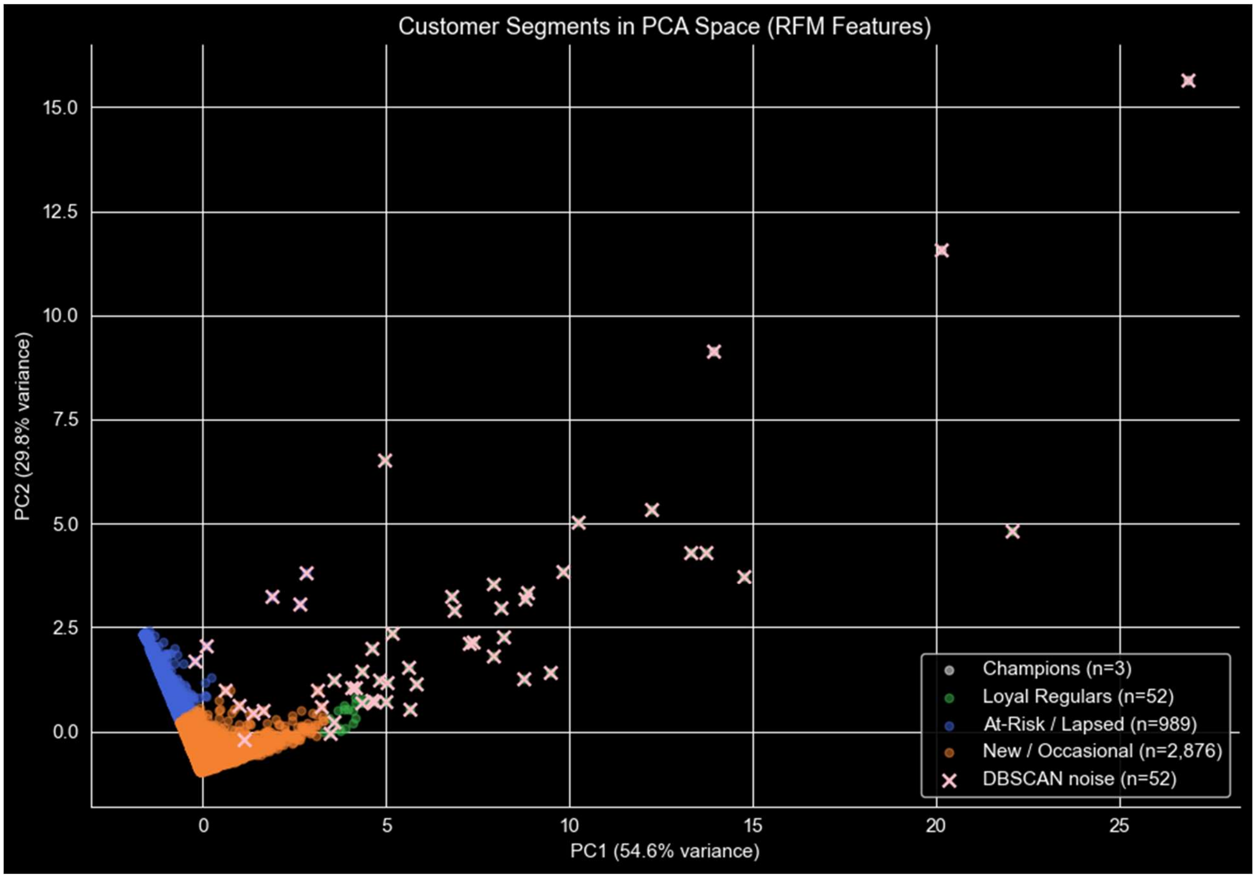


Figure 4 PCA space showing total variance captured as well as DBSCAN noise.

- *Seasonal Analysis – customer segment – revenue – season crosstabulation.*

The following tables show revenue in GBP and invoice count cross tabulated against customer segments by season:

SEASON	AT-RISK / LAPSED	CHAMPIONS	LOYAL REGULARS	NEW / OCCASIONAL
WINTER	194995.0	240537.0	433785.0	899576.0
SPRING	210935.0	61386.0	308070.0	847935.0
SUMMER	115011.0	95905.0	361128.0	936937.0
AUTUMN	0.0	224853.0	517973.0	1859367.0

Table 8 Showing customer revenue per season.

SEASON	AT-RISK / LAPSED	CHAMPIONS	LOYAL REGULARS	NEW / OCCASIONAL
WINTER	608	16	571	2572
SPRING	651	19	558	2410
SUMMER	315	19	531	2717
AUTUMN	0	54	617	4988

Table 9 Showing customer invoices per season.

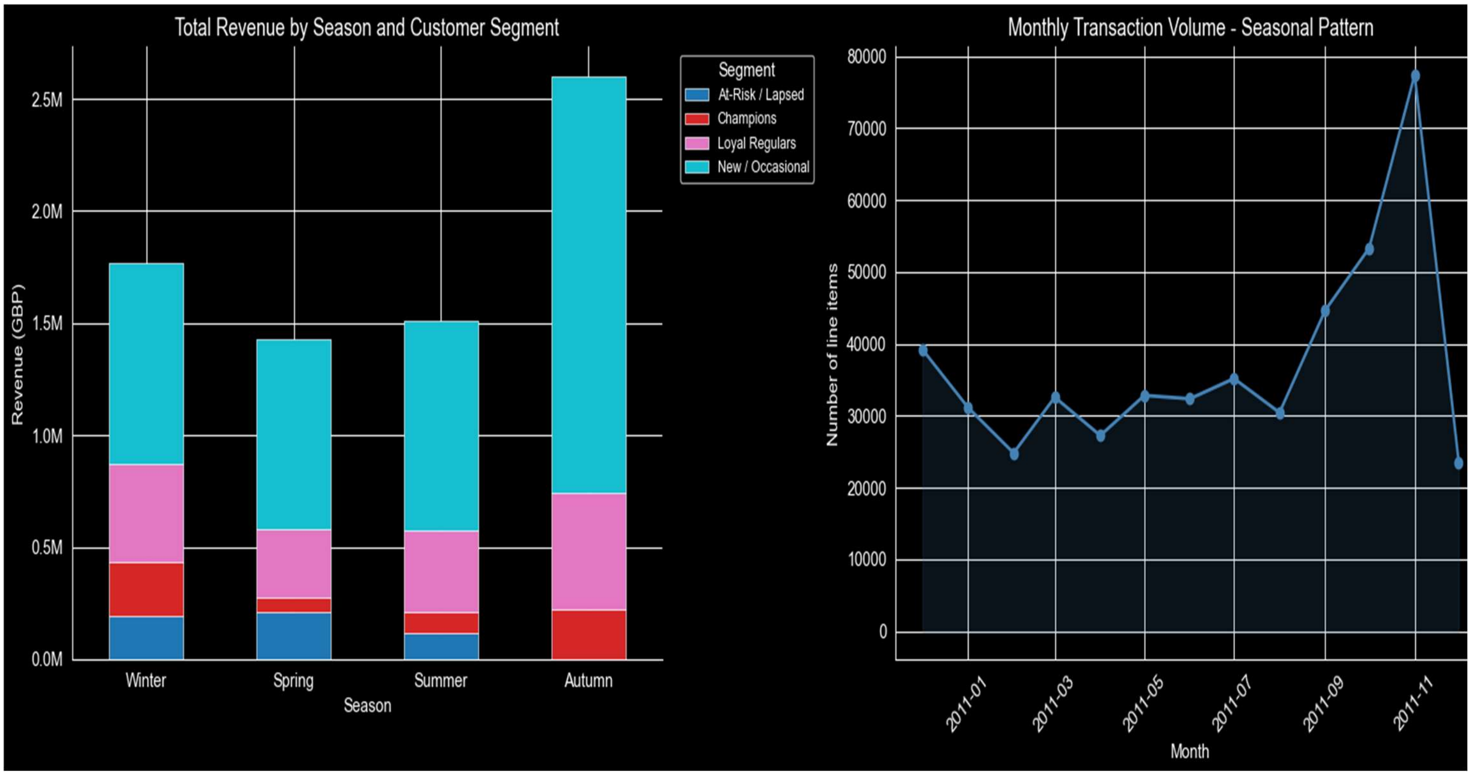


Figure 5 On the left: graph of Season vs Revenue. On the right: graph of Month v Transaction Volume

INTERPRETATION.

- *Revisiting Discovery Question.*

DISCOVERY QUESTION	ANSWERED?	ANSWER	CAVEATS.
WHAT PRODUCTS ARE FREQUENTLY PURCHASED TOGETHER?	Yes – fully	3 co-purchasing families.	Rules are generated only from UK invoices, which cannot be used for international inferences.
WHAT NATURAL CUSTOMER SEGMENTS EXIST IN THE DATA?	Yes – fully	4 segments: Champions(n=3), Loyal Regulars (n=52), At-Risk /Lapsed(n=989), New/Occasional (n=2876)	Segment labels are arbitrarily assigned, Champions are a micro cluster.
WHAT DISTINCT PURCHASING PATTERNS EMERGE BY SEASON?	Yes – fully	Universal purchasing peak among customer segments except At-	Single year data prevents generalizing the seasonal patterns.

	Risk customers during the Autumn season; 50% of the champions' purchases occur in the autumn season. At-Risk customers do not have any transactions during the Autumn season.

- *Domain Level Interpretation.*

  - The herb markers of lift 71.24 leads the findings from the association rules results. These items are basically sold as an inseparable set since no customer buys one variety without buying the others. Bundling these as a single curated product would greatly reduce decision fatigue for customers and increase average order or basket values per customer shopping. The Jumbo Bag Collection with a lift of 9.53 and Regency Teacup Set with lift of 26.75 should also be co-displayed and grouped to achieve a similar result.
  - With a heavily skewed customer base composed of mainly new/Occasional customers (73.4%), it would be important to tap into this base to convert a portion of them to loyal regulars generating more consistent revenue. The At-Risk/Lapsed customer segment (25.2%) characterized by their average recency of 244.6 days should be targeted with Win-Back campaigns during the Winter and Spring seasons where they are at least present and active.
  - The three champions account for a disproportionate amount of the Autumn revenue (£224,853 in a single season). These customers average a mean spend of £207,560 between themselves with mean frequencies of 36 invoices. This indicates that these three customers could be bulk buyers or wholesalers who may warrant individual and personalized relationships with the retailer as opposed to a regular relationship.

CRITICAL ASSESSMENT.

To show the validity of the results, the following robustness checks were performed:

- K-Means was re-run with five random seeds (0,7,13,42,99). These all returned a cluster size of 3 for the champions with stable silhouette scores of 0.595 across all seed values.
- Hierarchical clustering independently reproduced a four-cluster structure comparable to the four K-Means centroids, thus serving as a cross-validation method.
- Both Apriori and FP-Growth produced identical rule sets; 177 rules, indicating that there was no algorithmic bias.
- The decision tree was evaluated with a 5-fold stratified cross-validation with a max depth of 4 to balance interpretability and generalization.

- PCA component loadings confirm that principal component one and principal component two, PC1 & PC2 together capture 84.4% of the RFM variance. This shows that the two-dimensional visualizations capture most of the important aspects of the data.

Although the findings were substantial, there were some limitations that accompanied the data.

- The data only covered one year and as such any seasonal findings should not be considered as facts to inform generalizations over other time periods.
- The precision of the loyal regulars from the Decision Tree and Naïve Bayes: 0.395 and 0.406 respectively are low enough to assume that these loyal regulars may belong to other segments since they share certain overlapping RFM profiles with the larger customer base.
- The segment names are business interpretive labels given to the K-Means cluster IDs. These labels are also based on how well RFM reproduces the clustering and not how accurately real customer types are identified.
- RFM analysis treats all products and transactions uniformly so a customer who bought multiple items can have the RFM profile of one who bought one item. It also does not consider factors such as product categories and customer demographics.
- Customers without valid customer IDs were removed from the dataset. These could be anonymous customers or customers without store accounts.

Some ethical considerations which can be used as light posts to determine how the information resulting from the analysis is used can include:

- When considering tiered loyalty programs, these should apply fairly to all customer segments and comply with any existing regulations. These programs should not seek to maximize revenue at the expense of the customer regardless of the segment they appear to belong to.
- Due to the obvious purchasing peak that occurs during the Autumn season, any use of this information to facilitate artificial scarcity should be refrained from.
- Customers such as customer 18102 who appears to be a top anomalous customer could be targeted by unscrupulous individuals should any detailed anomaly assessing information get into the wrong hands. As such, any information concerning anomalous customers should be only shared using visualization methods that provide a wholistic outlook on such information.

CONCLUSION.

The analysis demonstrated that data mining techniques can generate meaningful and actionable business insights on retail transaction data by combining association rule mining, RFM based customer segmentation, classification, anomaly detection, and seasonal analysis.

The results revealed three major co-purchase product families, four business interpretable customer segments, and a purchasing behavior among the customer segments throughout meteorological seasons. These findings suggest practical business actions such as bundling co-purchase products, customer retention strategies, and seasonal planning.

However, these findings should be interpreted with caution as the data covers a limited period and lacks demographic considerations. Overall, all discovery questions were effectively answered and validated with some room for more future analysis.